

**Department of Computer Engineering**  
**Academic Term: July-November 2022**

**Class : B.E. Computer**  
**Semester : VII**

**Subject Name :ML**  
**Subject Code : CSC701**

|                             |                                            |
|-----------------------------|--------------------------------------------|
| <b>Practical No:</b>        | 7                                          |
| <b>Title.</b>               | To Study and implement Ensemble Techniques |
| <b>Date of Performance.</b> | 19/09/2025                                 |
| <b>Roll No:</b>             | 9913                                       |
| <b>Name of the Student:</b> | Mark Lopes                                 |

**Evaluation**

| <b>Indicator</b>           | <b>Very Poor</b>                  | <b>Poor</b>                       | <b>Average</b>        | <b>Good</b>                          | <b>Excellent</b>          |
|----------------------------|-----------------------------------|-----------------------------------|-----------------------|--------------------------------------|---------------------------|
| <b>Timeline (2)</b>        | More than three sessions late (0) | More than two sessions late (0.5) | Two sessions late (1) | One session late (1.5)               | Early or ontime (2)       |
| <b>Efforts(4)</b>          | N/A                               | N/A                               | Not Completed (1)     | Partially Completed (2)              | Completed(4)              |
| <b>Legibility(2)</b>       | N/A                               | N/A                               | Poor(1)               | Good(1.5)                            | Very Good(2)              |
| <b>Oral Assessment (2)</b> | N/A                               | N/A                               | N/A                   | Partially Understood the concept (1) | Understood the concept(2) |

```
In [ ]:import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
In [ ]:dataset = pd.read_csv("/content/PlayTennis.csv")
```

```
In [ ]:mapping = {"Sunny":1,"Overcast":2,"Rain":3} dataset
= dataset.replace({"Play Tennis":mapping}) mapping =
{"Hot":1,"Mild":2,"Cool":3}
dataset = dataset.replace({"Play Tennis":mapping})
mapping = {"High":1,"Normal":2}
dataset = dataset.replace({"Play Tennis":mapping})
mapping = {"Strong":1,"Weak":2}
dataset = dataset.replace({"Play Tennis":mapping})
mapping = {"Yes":1,"No":2}
dataset = dataset.replace({"Play Tennis":mapping})
```

```
In [ ]:dataset.head(15)
```

```
Out[ ]:
Outlook Temperature Humidity
Wind Play Tennis 0 1 1 1 2 2 1 1 1 1
10 1 2 2 1 1 1 1 2 2 1 1 1 1 2 2 1 2 2 1
13 3 2 1 1 2
1 2 2 2 1 1 2 1 3 3 2 1 2 1 4 3 3 2 2 1
5 3 3 2 1 2 6 2 3 2 1 1 7 1 2 1 2 2 8 1
```

```
In [ ]:X = pd.DataFrame(dataset[['Outlook','Temperature','Humidity','Wind']])
Y = pd.DataFrame(dataset['Play Tennis'])
```

```
In [ ]:from sklearn.model_selection import train_test_split
```

```
In [ ]:X_train,X_test,Y_train,Y_test = train_test_split(X,Y,test_size=0.2)
```

```
In [ ]:from sklearn.tree import DecisionTreeRegressor
regressor = DecisionTreeRegressor(criterion="mse",random_state=100,max_depth=4,min_
```

```
In [ ]: regressor.fit(X_train,Y_train)
```

Criterion 'mse' was deprecated in v1.0 and will be removed in version 1.2. Use `criterion='squared\_error'` which is equivalent.

```
Out[ ]:
/usr/local/lib/python3.7/dist-packages/sklea DecisionTreeRegressor(criterion='mse',
rn/tree/_classes.py:363: FutureWarnin g: max_depth=4, random_state=100)
```

```
In [ ]:from sklearn.tree import export_graphviz
import pydotplus
```

```
export_graphviz(regressor,out_file = "/content/model.dot")
```

```
In [ ]:y_pred = regressor.predict(X_test)
```

```
In [ ]:y_pred
array([2.,
2., 1.])
Out[ ]:
```

```
In [ ]:Y_test["Play Tennis"]
```

```
10 1
```

```
Out[ ]:      Name: Play Tennis,
```

```
13 2      dtype: int64
```

```
1 2
```

```
In [ ]:from sklearn.metrics import accuracy_score,mean_squared_error
```

```
      accu = accuracy_score(y_pred,Y_test)
```

```
      mse = mean_squared_error(y_pred,Y_test)
```

```
      print("Accuracy",accu)
```

```
Accuracy 1.0
```